

The Largest Defensible Local Edge Family in the Frozen Declared Corpus

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

TPC-173 freezes and completely reviews the contiguous TPC-133–172 theorem corpus, finding zero qualifying local actual-occurrence edge claims. TPC-174 proves a finite witness-verifier interface, but its only positive fixture is synthetic L0 and its production witness is absent. We now form the largest defensible production local-edge family relative to these exact sources.

An admissible family member must resolve to a qualifying TPC-173 claim and belong to a production candidate accepted by TPC-174. Since the qualifying source universe is empty, the largest family is exactly empty. It covers zero of the 2988 frozen cut rows; all 2988 are unmatched. In particular the nonempty-local-fibre hypothesis of the TPC-165 gluing theorem is not instantiated. The conclusion is maximal only inside the frozen declared corpus. It is a scoped L1 obstruction, not mathematical nonexistence, not actual active support, not canonical minimality, and not arithmetic L2 progress.

1 Admissible universe

Let $\mathcal{Q}_{173}$ be the TPC-173 qualifying claim set and let $\text{Verify}_{174}(W)$ mean that a production-mode witness passes the TPC-174 verifier [2, 3].

Definition 1.1 (Defensible local edge). *A local edge e is defensible in the current evidence universe if:*

1. *its source claim belongs to $\mathcal{Q}_{173}$; and*
2. *it belongs to a production witness W with $\text{Verify}_{174}(W) = \text{PASS}$.*

The family $\mathcal{E}_{175}$ contains every such edge and no other row.

Synthetic fixtures, shadow rows, archive addresses, formal gluing hypotheses and actual-core arithmetic statements without a cut crosswalk are therefore excluded by definition.

2 Scoped empty-family theorem

Theorem 2.1 (Largest declared-corpus family). *For the source-locked TPC-173 and TPC-174 artifacts,*

$$|\mathcal{Q}_{173}| = 0, \quad |\mathcal{E}_{175}| = 0. \quad (1)$$

Moreover $\mathcal{E}_{175} = \emptyset$ is the unique largest defensible family in the evidence universe of [definition 1.1](#).

Proof. TPC-173 has an empty qualifying claim list. Every defensible edge must map to one member of that list. Therefore no edge satisfies the first predicate in [definition 1.1](#), independently of the second. The family of all admissible members is consequently empty and contains every other admissible family, proving maximality in the declared universe. $\square$

Proposition 2.2 (Closed-world boundary). *Equation (1) proves neither mathematical nonexistence of local occurrences nor emptiness after a declared corpus enlargement.*

Proof. The admissible universe explicitly quantifies only over the finite TPC-133–172 corpus and the current committed production witness set. No completeness theorem for all mathematical sources is supplied. $\square$

3 Coverage and gluing state

The finite production archive has 2988 cut rows. Since $\mathcal{E}_{175}$ is empty,

Quantity	Count	
eligible source carriers	0	
local edges	0	
covered cuts	0	(2)
duplicated cuts	0	
unmatched cuts	2988	

TPC-165 assumes a nonempty local row family over every cut in a declared cover [1]. The empty family does not satisfy that hypothesis. Thus there is no production overlap cocycle, quotient, or formal global totality to compute. Vacuous set identities must not be reported as a production gluing pass.

4 H1 consequence and dynamic route

The local-edge root now has the sharper status

EMPTY_IN_FROZEN_DECLARED_CORPUS,

while the actual-active-support and canonical/minimal-representation roots remain independently not-testable. No actual-carrier impossibility or architecture stop follows.

The next coverage paper should record (2) and the noninstantiation of TPC-165. Without a sourced carrier, active-support and canonicity papers can only certify that their predicates remain not-testable. A later integration must retain all three roots unless the corpus is explicitly enlarged or new source mathematics is supplied.

The executable validator fixes the complete claim boundary, coverage and gluing state, scoped maximality certificate, and H1 architecture state. Independent mutations flip every positive claim field and separately test formal global totality, enlarged-corpus nonexistence, and architecture-stop promotion.

5 Claim firewall

This is a scoped L1 empty-inventory/maximality theorem with finite L0 checking. It proves no nonempty production local family, formal global totality, actual active support, canonical/minimal actual representation, fixed- h_0 edge theorem, named fixed phase, positive fixed- X L2, strict 1/400, prime-pair lower bound, or twin-prime theorem.

References

- [1] Liang Wang. Source-backed local-to-global gluing for formal h1 occurrence crosswalks. *TPC Research Archive*, 2026. TPC-165.
- [2] Liang Wang. Production source-claim inventory. *TPC Research Archive*, 2026. TPC-173.
- [3] Liang Wang. A minimal source-locked local occurrence-edge witness contract. *TPC Research Archive*, 2026. TPC-174.